Top ten real world problem statements:

Domain 1: Computing (AI, Data Science, Cybersecurity)

Problem 1: Real-time Predictive Gender Safety Analytics

- **The Challenge:** Develop Edge-AI software for CCTV that detects "high-risk" scenarios (e.g., a lone woman surrounded by a large group in an unlit area).

Problem 2: AI-Powered Acoustic Sniper Localization

- **The Challenge:** Build a microphone array system using **Deep Learning** to filter out battlefield noise (explosions, wind) and pinpoint a shooter's coordinates on a digital map in under 500ms.

Domain 2: Electrical and Electronics (IoT, Power, Semiconductors)

Problem 3: Self-Learning Appliance Resource Optimizer

The Challenge: Design a low-cost IoT module that can be retrofitted onto *existing* appliances to monitor vibration/current and optimize cycles for 20% water/energy saving.

Problem 4: Non-Intrusive Load Monitoring (NILM) for EV/Solar

The Challenge: Use High-Frequency Sampling of the single point of entry (the meter) to identify the "electrical fingerprint" of an EV charger or solar inverter without entering the house.

Domain 3: Mechanical and Energy (Robotics, Manufacturing, CCU)

Problem 5: Electrostatic Dust Suppression for Mining

- **The Challenge:** Develop an automated drone or mobile robot equipped with **electrostatic sprayers** that bond to dust particles in open-pit mines, using 80% less water than traditional tankers.

Problem 6: Modular Carbon Capture for Small-Scale Boilers

The Challenge: Design a portable "Plug-and-Play" CCU unit for small-scale industrial boilers (common in MSMEs) that converts flue gas into liquid for greenhouse use.

Domain 4: Bio, Healthcare, and Agriculture

Problem 7: Graphene/Bio-Char Based PFAS Remediation

The Challenge: Develop a filtration cartridge using **locally sourced agricultural bio-char** treated with nanoparticles to selectively bind and neutralize PFAS in groundwater.

Problem 8: Frugal "Micro-Milling" Units for Farmer Collectives

- **The Challenge:** Design a solar-powered, pedal-hybrid grain husker/miller that can be maintained by a local blacksmith. Must reduce manual labor by 70%.

Domain 5: Civil and Environmental (Infrastructure, Smart Cities)

Problem 9: AI-Driven Ultrasonic Scanner for RAAC Detection

- **The Challenge:** Create a handheld "Structural Health Scanner" that uses Ultrasonic Pulse Velocity (UPV) data fed into a mobile app to instantly flag the presence of RAAC in aging public buildings.

Problem 10: Digital Twin for Urban Flash-Flood Mitigation

- **The Challenge:** Combine IoT water-level sensors in city drains with a **BIM (Building Information Modeling)** map to predict exactly which street will flood 30 minutes before it happens.

Real World Problem Statements form VIT Estates office:

1. **Steam production by refurbishing the existing Solar dish near Mens Hostel office STP – waste water treatment with steam or heat from the solar dish – steam or heat usage at hydrogen production from biogas (Methane)**
2. **Low cost and low energy based Salt extraction system for the back wash / re charging of WTP and RO plants**
3. **Co2 absorption system for the bio gas based Generator and convert Co2 absorbed in to tar like material which may be used as tiles, or laying road**
4. **Fuel cell – microbes based working with gray water / STP water**
5. **Green hydrogen production from gray water – electrolysis**
6. **Heat recovery system for VRF /VRV**
7. **Chillness recovery from Heat pump or combine both the problems of heat pump and VRF/ VRV**
8. **VIT campus centralised water grid monitoring – consumption of each building, water extraction of each source**
9. **Centralised energy monitoring for VIT campus**
10. **Coating materials for metals/ metal structure which resist the rusting at WTPs and STPs**
11. **DC microgrid system with solar PV in the roof top and DC appliances at any one floor of the hostel or small buildings**
12. **Particulate material reduction system for DGs – best alternatives of Retrofit emission control devices RECD**
13. **Smart dust bin – waste segregation at least three types – recyclables (plastics, papers etc) , non-recyclables, organic wastes**
14. **Low cost biomass digester for the campus biomass, food wastes etc**
15. **Bio gas purification plant – washing of bio gas**
16. **Automatic salt loading setup for WTP /RO plants**

17. **Extraction and handling of oxygen from Electrolysis process to use it in STP for enhancing the operation like Cod , BOD improvement**
18. **Business / commercial model for selling Vermi composte / mulch bed / STP manure**
19. **Low cost STP sludge drier and convert into power form**
20. **Alternative or green water treatment methods for softening / treating waste water**
21. **Low cost High voltage pulse application for gray water treatment- killing microbes- an alternative of UV light**
22. **Garbage dust bin automation – mobile garbage bins**
23. **Decomposition or conversion single use plastics**
24. **Modern / automatic gates to restrict stray dog entries**
25. **Feasibility study to arrive optimal solutions for making VIT vellore campus as Net zero energy campus – carbon neutral campus**
26. **E bike – Low cost / optimal conversion of old bi cycles of VIT campus into E bike**
27. **Low cost Air quality monitoring for VIT campus**
28. **Automatic Corridor floor cleaning and mobbing robot / machine**
29. **Smart Parking slot availability indicator for PRP phase 2 basement**
30. **Occupancy sensor based appliance control for class rooms, corridors, rest rooms street lights (light dimmer or alternate lights ON when no body moves)**